

ADVANCED SQL : DATE FUNCTIONS

EXERCISE 6

Table 1: Employees

emp-id	name	hire-date
1	Alice	2020-01-15
2	Bob	2021-06-10
3	Charlie	2023-03-22

Q1. `SELECT emp-id,
name,
hire-date,
DATEADD(MONTH, 6, hire-date) AS hire-plus-6-months
FROM Employees;`

emp-id	name	hire-date	hire-plus-6-months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

Table 2: Students

student-id	name	dob
101	Maya	2005-08-10
102	Ethan	2004-12-01
103	Sienna	2006-03-15

Q2. `SELECT student-id,
name,
DATEDIFF(DAY, dob, '2026-05-04') AS age-in-days
FROM Students;`

student-id	name	gpc-in-days
101	Maya	7572
102	Ethan	7824
103	Sienna	7355

Table 3: Events

event-id	event-name	event-date
1	Seminar	2024-06-15
2	Workshop	2025-09-01
3	Hackathon	2025-01-20

Q3. `SELECT event-id,`
`event-name,`
`DATEIFF(DAY, '2026-05-04', event-date) AS days-remaining`
`FROM Events;`

event-id	event-name	days-remaining
1	Seminar	-688
2	Workshop	-245
3	Hackathon	-469

Table 4: Invoices

invoice-id	issue-date	due-date
501	2025-03-10	2025-03-25
502	2025-04-01	2025-04-15
503	2025-04-10	2025-04-20

Q4. `SELECT invoice-id,
 issue-date,
 due-date,
 DATEDIFF(DAY, issue-date, due-date) AS days-between
 FROM Invoices;`

invoice-id	issue-date	due-date	days-between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

Table 5: Courses

course-id	name	start-date
201	SQL Basics	2025-05-01
202	Python	2025-06-01

Q5. `SELECT course-id,
 name,
 TO-CHAR(start-date, 'Month YYYY') AS formatted-date
 FROM Courses;`

course-id	name	formatted-date
201	SQL Basics	May 2025
202	Python	June 2025

Table 6: Memberships

membership-id	start-year	start-month	start-day
1	2023	5	10
2	2022	11	25

Q6. `SELECT membership-id,` `AS full-start-date`
`DATEFROMPARTS (start-year, start-month, start-day)`
`FROM Memberships;`

membership-id	full-start-date
1	2023-05-10
2	2022-11-25

Table 7: Subscriptions

sub-id	plan	renewal-date
11	Basic	2025-01-01
12	Premium	2025-03-15

Q7. `SELECT sub-id,`
`plan,`
`DATEADD (YEAR, 1, renewal-date) AS extended-renewal-date`
`FROM Subscriptions;`

sub-id	plan	extended-renewal-date
11	Basic	2026-01-01
12	Premium	2026-03-15

Table 8: Orders

order-id	order-date
1001	2025-04-15
1002	2025-04-10

Q8.
 SELECT order-id,
 order-date,
 CURRENT-DATE AS today-date,
 DATEDIFF(DAY, order-date, CURRENT-DATE) AS days-since-order
 FROM Orders;

order-id	order-date	t=day-date	days-since-order
1001	2025-04-15	2026-05-04	384
1002	2025-04-10	2026-05-04	389

Table 9: Trainings

training-id	topic	training-date
1	Safety	2025-01-10
2	Compliance	2025-02-20

Q9.
 SELECT training-id,
 topic,
 DATE-PART('year', training-date) AS training-year
 FROM Trainings;

training-id	topic	training-year
1	Safety	2025
2	Compliance	2025

Table 10: Blog-Posts

post-id	title	published-on
1	SQL Tips	2025-04-01 10:15:00
2	Data Cleaning	2025-04-12 16:45:00

Q10. `SELECT post-id,`
`title,`
`EXTRACT (HOUR FROM published-on) AS hour-published,`
`EXTRACT (MINUTE FROM published-on) AS minute-published`
`FROM Blog-Posts;`

post-id	title	hour-published	minute-published
1	SQL Tips	10	15
2	Data Cleaning	16	45

Table 11: Drivers

driver-id	license-expiry
301	2025-08-10
302	2023-12-31

Q11. `SELECT driver-id,`
`license-expiry,`
`DATEDIFF (DAY, CURRENT-DATE, license-expiry) AS days-left`
`FROM Drivers;`

driver-id	license-expiry	days-left
301	2025-08-10	-267
302	2023-12-31	-855

Table 12 : Messages

message-id	sent-timestamp
1	2025-04-19 09:32:45
2	2025-04-18 23:59:59

Q12.
 SELECT message-id,
 sent-timestamp,
 CURRENT-TIMESTAMP AS current-timestamp,
 DATEDIFF(SECOND, sent-timestamp, CURRENT-TIMESTAMP)
 FROM Messages; AS seconds-since-sent

message-id	sent-timestamp	current-timestamp	seconds-since-sent
1	2025-04-19 09:32:45	2026-05-04 00:00:00	32 020 035
2	2025-04-18 23:59:59	2026-05-04 00:00:00	32 054 401

Table 13 : Returns

return-id	return-date
901	2025-04-05
902	2025-04-01

Q13.
 SELECT return-id,
 return-date,
 DATEADD(DAY, 15, return-date) AS restock-date
 FROM Returns;

return-id	return-date	restock-date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

Table 14 : Assignments

assigned - 1st	assigned - on
1	2025-03-01
2	2025-03-05

Q.14. `SELECT assign-id, assigned-on-date
TO-DATE (assigned-on, 'YY-MM-DD') AS
FROM Assignments;`

assign-id	assigned-on-date
1	2025-03-01
2	2025-03-05

Table 15: Meetings

meeting-id	scheduled-time
1	2025-04-19 14:00:00
2	2025-04-19 09:30:00

Q15. `SELECT meeting-id,
TO-CHAR(scheduled-time, 'Month DD, YYYY "at" HH12:MI AM')
FROM Meetings;` AS formatted-meeting-time

meeting-id	formatted-meeting-time
1	April 19, 2025 at 02:00 PM
2	April 19, 2025 at 09:30 AM